

DATE	TEAM ID	PROJECT NAME	MAXIMUM MARKS
1.10.2026	SWTID-2026-8437	Power BI Analytics	[Evaluation Pending]

1. PROJECT OVERVIEW & OBJECTIVE

This project delivers an enterprise Power BI Analytics Dashboard integrated with automated ticket classification flow design. The dashboard incorporates dynamic Row-Level Security (RLS) driven by script-controlled access-control rules. Users in the **EEE branch** can view ticket metrics where Branch = EEE, non-EEE users are restricted from viewing protected EEE analytics, and **Administrators** maintain complete reporting visibility across all branches.

2. SETUP & EXECUTION PIPELINE STEPS

Step	Pipeline Stage	Technical Description	Screenshot / Artifact
1	Data Source Connection	Import ticket dataset into Power BI Desktop via direct connector / REST endpoint.	[Data Source Config]
2	Power Query ETL	Cleanse Branch field, remove nulls, set data types, and apply transformations.	[Power Query Editor]
3	Data Modeling	Construct Star Schema connecting Fact_Tickets with Dim_User & Dim_Branch.	[Model View Schema]
4	DAX Metrics Engine	Author key measures: Total Tickets, Resolution Rate, and Branch Authorization filters.	[DAX Measure Table]
5	RLS Role Definition	Configure DAX filter rule [Branch] = USERPRINCIPALNAME() or EEE role logic.	[RLS Role Manager]

3. STAR-SCHEMA DATA MODEL ARCHITECTURE

Table Name	Table Type	Key Attributes	Relationship / Cardinality
Fact_Tickets	Fact Table	TicketID, UserID, Branch, Status, Priority, CreatedDate	N/A (Central Fact)
Dim_User	Dimension Table	UserID, UserName, UserEmail, AssignedBranch, Role	1-to-Many on User ID
Dim_Branch	Dimension Table	BranchID, Branch, Location, RegionalHead	1-to-Many on Branch
Dim_Calendar	Date Dimension	DateKey, Date, Year, Month, Quarter	1-to-Many on CreatedDate

4. PROJECT-SPECIFIC ACCEPTANCE & RLS VALIDATION

Test Scenario	Expected Result	Actual Result	Status
EEE user views dashboard	Renders metrics for Branch = EEE records only	To be filled	☐
Non-EEE user views dashboard	EEE branch ticket data is completely restricted/hidden	To be filled	☐
Administrator views dashboard	Renders global metrics across all branches	To be filled	☐
Empty / Missing Branch field record	Follows secure RLS fallback rule (Hidden by default)	To be filled	☐

5. EVIDENCE CHECKLIST

- ✓ Power Query data transformation and Branch field cleansing screenshot.
- ✓ Power BI Relationship Data Model (Star Schema) view screenshot.
- ✓ DAX Measure definitions and Row-Level Security (RLS) script rule screenshot.
- ✓ Authorized EEE user filtered dashboard report view screenshot.
- ✓ Restricted Non-EEE user report view / access-denied screenshot.
- ✓ Administrator global unfiltered dashboard report view screenshot.

6. STAKEHOLDER SIGN-OFF & EVALUATION

Role	Name	Date	Signature
Developer / Tester	_____	_____	_____
Project Lead / Guide	_____	_____	_____
Product Owner / Evaluator	_____	_____	_____